

MATH 345 Skeleton Notes

Unit 5: Optimization and training

Section 5.5: Hessians and the second derivative test

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The two-variable second derivative test

As in single-variable calculus, a critical point of a function is not necessarily a local extremum. This happens at a *saddle point*.

Definition A critical point of a function which is not a local minimum or local maximum is called a saddle point.

The *second derivative test* provides a method for determining whether a local extremum occurs at a critical point of a function. The test is more complicated for two-variable functions than the second derivative test for functions of a single variable, given that there are *four* second derivatives rather than just one derivative.

Theorem: The Two Variable Second Derivative Test Suppose $f(x, y)$ is a scalar-function of two variables, and the first and second derivatives of f are continuous on a disk containing some critical point (a, b) for f . Define the discriminant

$$D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - f_{xy}(a, b)^2$$

Then:

- If $D(a, b) > 0$ and $f_{xx}(a, b) > 0$, then (a, b) is a *local minimum* of f .

- If $D(a, b) > 0$ and $f_{xx}(a, b) < 0$, then (a, b) is a *local maximum* of f .
- If $D(a, b) < 0$, then (a, b) is a *saddle point*.
- If $D(a, b) = 0$, then the test does not tell us whether (a, b) is a local extremum.

Remark From our current perspective, the second derivative test is quite mysterious. It seems unclear why the discriminant D is able to detect local extrema. We will later see in Activity act-10-1-second-derivative-test-two-variables, using our knowledge of linear algebra from Optimization and training, in particular the theory of eigenvalues, *why* the second derivative test detects local extrema, and also determine an analogue of the second derivative test for functions of more than two variables.

Activity

Use the second derivative test to determine if the critical point $(1, -2)$ of $f(x, y) = x^2 + y^2 - 2x + 4y$ found earlier is a local maximum, a local minimum, or a saddle point.

Activity

Use the second derivative test to analyze the critical point $(0, 0)$ of $f(x, y) = x^2 - y^2$ found earlier.

Activity

Find all local maxima, local minima, and saddle points for the function

$$f(x, y) = x^2 + xy + y^2 - 6x + 6.$$

In the previous examples, the discriminant of the functions were constant because f was a quadratic function. Let us consider an example where the discriminant is non-constant.

Activity

Find all local maxima, local minima, and saddle points for the function $f(x, y) = x^3 + y^2 + 2xy - 4x - 3y + 5$.

Remark The following activity explains why we only look at f_{xx} in the first two cases of the second derivative test.

Activity

True or False: If $D > 0$, then f_{xx} and f_{yy} have the same sign.

Saddle points

Note A common way to detect a saddle point is to find two directions through the point such that the function increases in one direction and decreases in another. This is not a complete definition of every possible saddle point; it is a useful test in many examples.

For example, the function

$$f(x, y) = x^2 - y^2$$

has a saddle point at $(0, 0)$. Along the line $y = 0$,

$$f(x, 0) = x^2,$$

which has a local minimum at $x = 0$. Along the line $x = 0$,

$$f(0, y) = -y^2,$$

which has a local maximum at $y = 0$. The contour plot below shows the same behavior.

In this example, the surface increases in one coordinate direction and decreases in the other. This is the behavior suggested by the saddle shape.

Remark More formally, if the lines are given in parametric form by

$$\{\mathbf{a} + t\mathbf{d}_1\} \quad \text{and} \quad \{\mathbf{a} + t\mathbf{d}_2\},$$

then the one-variable restrictions

$$g(t) = f(\mathbf{a} + t\mathbf{d}_1) \quad \text{and} \quad h(t) = f(\mathbf{a} + t\mathbf{d}_2)$$

can show opposite local behavior at $t = 0$. In the example above, this verifies that nearby points occur on both sides of $f(\mathbf{a})$.

Activity

For each point in the figure below, indicate whether it is a local maximum, a local minimum, or a saddle point.

The second derivative test

In this subsection, we now obtain a form of the second derivative test for scalar-valued functions with more than two variables.

Quadratic approximation and quadratic forms

Recall that if a single-variable function f has n derivatives at $x = a$, then the n th Taylor polynomial for f at a is the function

$$\begin{aligned}
 p_n(x) &= f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n \\
 &= \sum_{k=0}^n \frac{f^{(k)}(a)}{k!}(x - a)^k.
 \end{aligned}$$

In particular, for $n = 1$ we obtain the *linear approximation*

$$p_1(x) = f(a) + f'(a)(x - a)$$

of f , and for $n = 2$ we obtain the *quadratic approximation*

$$p_2(x) = f(a) + f'(a)(x - a) + f''(a)\frac{(x - a)^2}{2}.$$

The Taylor polynomial $p_n(x)$ approximates the function $f(x)$ near $x = a$, with the approximation improving as n increases.

Activity

Consider the function $f(x) = \sqrt[3]{x}$.

(a) Find the first and second Taylor polynomials for f at $x = 8$.

Activity

(b) Use these two polynomials to estimate $\sqrt[3]{11}$.

If a is a critical point of f , then the quadratic approximation of f is given by the expression

$$f(a) + f''(a)\frac{(x - a)^2}{2}$$

If $f''(a) > 0$, then the quadratic approximation of f is an upward pointing parabola with a vertex at $(a, f(a))$. This approximation has a minimum at this vertex, explaining a is a local minimum of the function f (since f is closely approximated by its quadratic approximation). Similarly, if $f''(a) < 0$, then the quadratic approximation is a downward pointing parabola with a maximum at $(a, f(a))$, explaining why a is a local maximum of the function f .

This explanation indicates how the second derivative test might work for a multi variable function. We obtain an analogue of the quadratic approximation of such a function. Then, if that quadratic approximation has a maximum or minimum at the critical point, then the actual function has a local minimum or maximum near that point. The quadratic approximation of the function f is defined in terms of the *Hessian* of f (recall Definition def-8-5-hessian).

Definition: Quadratic approximation The

quadratic approximation

$Q(\mathbf{x})$ of a scalar-valued function f near a point $\mathbf{a}$ is the function defined by

$$Q(\mathbf{x}) = f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a}) + \frac{1}{2}(\mathbf{x} - \mathbf{a})^T H_f(\mathbf{a})(\mathbf{x} - \mathbf{a})..$$

Remark The quadratic approximation of a function f refines the linear approximation via the gradient that we previously encountered. In particular, if the second derivatives of f are continuous, then

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} \frac{f(\mathbf{x}) - Q(\mathbf{x})}{\|\mathbf{x} - \mathbf{a}\|^2} = 0.$$

If $\mathbf{x}$ is close to $\mathbf{a}$, then $\|\mathbf{x} - \mathbf{a}\|^2$ is much smaller than $\|\mathbf{x} - \mathbf{a}\|$, so that the quadratic approximation $Q(\mathbf{x})$ to f is in general a much better approximation than the linear approximation $L(\mathbf{x})$.

Activity

Compute the gradient and the Hessian of the function $f(x, y) = \ln(xy - 1)$ at the point $(x, y) = (1, 2)$, use these values to determine the quadratic approximation to f at the point $(1, 2)$, and then use the quadratic approximation to estimate the value of f at $(1.2, 1.8)$.

Activity

Let $f(x, y) = \cos(x/y)$. Use the gradient and Hessian to obtain a quadratic approximation of the function $f(x, y)$ near $\mathbf{a} = (0, 1)$.

If $\mathbf{a}$ is a critical point of a function f , then the quadratic approximation of the function f takes the form

$$Q(\mathbf{x}) = f(\mathbf{a}) + \frac{1}{2}(\mathbf{x} - \mathbf{a})^T H_f(\mathbf{a})(\mathbf{x} - \mathbf{a}),$$

and so it suffices to understand the

quadratic form

associated with the Hessian matrix.

The second derivative test

Theorem: Second Derivative Test Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a function, and suppose $\mathbf{a} \in \mathbb{R}^n$ is a critical point of f .

1. If the Hessian $H_f(\mathbf{a})$ is positive definite then $\mathbf{a}$ is a local minimum of f .
2. If the Hessian $H_f(\mathbf{a})$ is negative definite then $\mathbf{a}$ is a local maximum of f .
3. If the Hessian $H_f(\mathbf{a})$ is indefinite then $\mathbf{a}$ is a saddle point of f .

In any other cases, the test gives no information.

Proof At a critical point, the first-order term in the Taylor approximation is zero. For small $\mathbf{h}$,

$$f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) = \frac{1}{2}\mathbf{h}^T H_f(\mathbf{a})\mathbf{h} + \text{terms smaller than } \|\mathbf{h}\|^2.$$

If $H_f(\mathbf{a})$ is positive definite, then the quadratic term is positive for every nonzero sufficiently small $\mathbf{h}$, so $f(\mathbf{a} + \mathbf{h}) > f(\mathbf{a})$ for all sufficiently small nonzero $\mathbf{h}$. This gives a local minimum. The negative definite case gives a local maximum. If the Hessian is indefinite, there are directions in which the quadratic term is positive and directions in which it is negative, so nearby points occur on both sides of $f(\mathbf{a})$. This gives a saddle point.

Activity: Predicting from Hessian eigenvalues

Suppose $\mathbf{a}$ is a critical point of a scalar-valued function f . For each list of eigenvalues of $H_f(\mathbf{a})$, classify the critical point as a local minimum, local maximum, saddle point, or inconclusive.

1. 3, 5
2. $-2, -7$
3. 4, -1
4. 0, 2

Tags. [U5-L05 | C+R | Core]

Activity

Show that a matrix

$$A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$$

is positive definite if and only if $ad - b^2$ is positive and $a > 0$, negative definite if and only if $ad - b^2$ is positive and $a < 0$, and indefinite if and only if $ad - b^2$ is negative. It follows from this that the second derivative test in two variables is just a special case of the second derivative test above.

Activity

Consider the function $f(x, y, z) = x^2 + y^2 - xz$. Find all critical points of f and determine the behavior of f near each critical point.

Task First find the critical points.

Task Now classify each critical point.

Activity

Classify the critical points of the function $f(x, y, z) = xy + yz + xz + xyz$ using the second derivative test.